

FarmTech MIS

Enterprise FPO & SHG Architectural Specification Guide

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Version: 2.1 (Production Release)

Date: May 2026

Target: Self-Help Groups (SHG), Farmer Producer Organizations (FPO)

Environment: Laravel 12 + MongoDB + Tailwind CSS

1. Executive Architectural System Summary

The **FarmTech MIS Portal** is a high-fidelity, dual-platform digital ecosystem specifically engineered to address the systemic challenges of agricultural fragmentation. By creating seamless structural pathways between individual farmers, aggregate cooperatives (Farmer Producer Organizations/FPOs), and micro-processing processing cells (Self-Help Groups/SHGs), the platform enables decentralized groups to leverage collective volume, scale up sourcing quality, and cut intermediate leakage.

This document serves as the absolute technical manual detailing the design paradigms, backend routes, database structures (utilizing high-performance hybrid setups of MySQL and MongoDB), and dynamic workflows that make up the FPO and SHG features of the system.

Core System Integration Strategy

Individual farmers input crop yields into **Crop Pools**, which are pooled village-wise. The FPO aggregates these pools into **Bulk Procurement Contracts** to wholesale them to large buyers. At the same time, SHGs buy these bulk raw ingredients directly from the FPO at pre-negotiated wholesale prices, processing them locally into high-margin consumer products.

2. FPO / FPG Dashboard Modules & Execution

The FPO dashboard operates as a robust Enterprise Resource Planning (ERP) center for agricultural aggregators. It manages 9 primary business features:

2.1 Farmer Management

Organizes, tracks, and audits individual farmer profiles village-wise. This includes recording landholding sizes, active crop types (such as Basmati Rice, Wheat, or Mustard), and tracking productivity scores using historical yield inputs.

2.2 Crop Pooling Hub

Allows cooperative managers to group smaller yields of grain collected from separate smallholders into standardized commercial batches (pools). This creates high-volume wholesale opportunities directly qualifying for Minimum Support Prices (MSP) or big-buyer tenders.

2.3 Logistics & Fleet Scheduling

Includes automatic routes mapping, truck dispatch registers, transport cost index calculations, and delivery schedules to minimize operational delay and overheads.

2.4 Equipment Custom Hiring

A digital booking directory for cooperative assets (Tractors, Seeders, Combine Harvesters). Smallholders can book high-capacity machinery by the hour or day directly from their local FPO center.

2.5 Proximity GIS Farmer Map

Uses vector map layers (via Leaflet and OpenStreetMap data) to graphically display crop fields, soil types, and cluster zones of member farmers within the district.

2.6 Yield & Income Forecast Engine

An analytical algorithm analyzing historical seasonal performance, water index tables, and crop categories to calculate FPO bulk production forecasts and projected revenues.

2.7 Marketplace B2B Orders

Tracks buyer purchase bids, wholesale procurement contracts, and processing shipments purchased by institutional distributors and Self-Help Groups (SHGs).

2.8 Government Compliance Reports Center

Generates fully compliant PDF summary audit sheets (Farmer Rosters, Scheme Penetration Reviews) directly adhering to Punjab Department of Agriculture guidelines. Every audit sheet is assigned a unique system hash code and saved directly into the MongoDB ledger.

3. Self-Help Group (SHG) Integration & Processing

Self-Help Groups are local women-led micro-enterprises. The FarmTech MIS equips SHGs with a suite of enterprise tools designed to run a secure, profitable, value-added business:

SHG Technical Infrastructure: Built with a brutalist-premium emerald palette, the SHG portal is structured using responsive single-page architectures (SPA) that display real-time inventories, local sourcing coordinates, and processing logs.

3.1 Bulk Sourcing Linkage to FPO

SHGs secure premium-grade raw materials (like mustard seeds, raw wheat, or turmeric) by initiating direct B2B purchases from the FPO. Sourcing directly from the FPO ensures fair pricing, guaranteed organic quality, and eliminates the middleman markup.

When an SHG confirms a bulk order, the system instantly logs a corresponding order within the FPO Marketplace Orders registry, linking the cooperatives seamlessly.

3.2 Local Food Processing & Packaging Logs

Provides a micro-ERP module to track processing workflows. For example, when mustard seed is cold-pressed into premium mustard oil, the system records:

- **Raw Sourced Inventory:** 50 Quintals Raw Mustard Seed.
- **Processing Input:** Milling & extraction logs with timestamps.
- **Consumer Goods Output:** 1,200 bottles (1L each) of packaged Premium Mustard Oil.
- **Batch Codification:** Automated traceability barcoding.

3.3 Proximity-Aware B2B Marketplace Finder

Enables SHGs to locate nearby buyers, retail distributors, and warehousing locations using coordinate-based radius queries. This utilizes spatial mathematics to identify nearby demand hubs within a 15-50km radius.

3.4 Proximity Coordinates & Automated WhatsApp Flows

To ensure reliable field communication, the FarmTech SHG platform is integrated with automated messaging alerts:

1. **Geographic Match:** Sourcing agent locates a processed packaging batch within proximity.
2. **API Hook Trigger:** A background webhook translates dispatch parameters.
3. **WhatsApp Dispatch Notification:** Dispatches automated WhatsApp template messages detailing weight, cost, pickup coordinates, and contact details to truckers and buyers.

3.5 Dynamic Training & Capacity Registration Portals

Provides skill advancement systems where SHG members register for agricultural processing workshops, packaging compliance tutorials, and government safety certification programs.

4. FarmTech Technology Stack Specification

The FarmTech MIS application employs a modular, secure, and modern tech stack designed to ensure peak operational reliability and lightning-fast user interactions:

Layer / Component	Technology Used	Operational Responsibility
Backend Framework	Laravel 12.0 / PHP 8.2+	Robust REST APIs, secure route groups, CSRF token validation, and MVC clean separation.
Database Engine	MongoDB v6.0+ (NoSQL)	Hybrid integration with MongoDB Laravel Driver for elastic storing of logs, user records, and dynamic audits.
Frontend Layout	Tailwind CSS / HTML5	Responsive layouts, clean modern components, and emerald color systems.
Interactivity Layer	Alpine.js / AJAX (Fetch API)	Single-page feeling, real-time dynamic dashboard tab swaps, and background submissions.
GIS Mapping	Leaflet.js / OpenStreetMap	Proximity calculations, vector coordinate markers, and interactive zoning maps.
PDF Engine	html2pdf.js / Dompdf PHP	Hybrid client-side rendering and backend backup engine to guarantee secure offline reporting.

4.1 Hybrid MongoDB Schemas & Architectural Models

Below is the exact schema representation utilized in the system for our primary dynamic components:

```
// ReportAuditLog Model Configuration
class ReportAuditLog extends Model {
    protected $connection = "mongodb";
    protected $collection = "report_audit_logs";
    protected $fillable = [
        "user_id", "hash_code", "report_type",
        "metrics", "authority", "status", "created_at"
    ];
}
```

```
// CropPool Model Configuration
class CropPool extends Model {
    protected $connection = "mongodb";
    protected $collection = "crop_pools";
    protected $fillable = [
        "farmer_name", "crop_type", "quantity",
        "unit", "price_per_unit", "total_value", "status"
    ];
}
```


5. Step-by-Step FPO & SHG Sourcing & Processing Flow

The absolute step-by-step workflow detailed below outlines how the system processes data from individual harvest up to processed retail commodity distribution.

Step 1: Supply Aggregation (Farmer ? FPO)

- **Action:** Individual farmers log their grains in their portal, or the FPO manager records them under the **Crop Pooling Hub**.
- **Database Entry:** A document is created in the `crop\_pools` collection in MongoDB with status `Pooled`.
- **Business Value:** Solves volume limits. Smaller batches are now officially part of a major commercial pooling lot.

Step 2: Wholesale B2B Purchase (SHG ? FPO Sourcing)

- **Action:** The SHG logs in and opens the raw sourcing inventory. They purchase raw wheat or mustard seeds in bulk from the pooled FPO stocks.
- **System Trigger:** An AJAX POST request is submitted. The backend automatically writes an entry in the `fpo\_orders` collection with status `Pending` (appearing instantly in the FPO Marketplace orders module).
- **Business Value:** Secures fresh, high-quality agricultural ingredients at wholesale pricing without intermediaries.

Step 3: FPO Order Approval & Dispatch (FPO ? SHG Logistics)

- **Action:** The FPO approves the bulk order in their portal, schedules a transport vehicle from the **Logistics & Fleet** list, and assigns a driver.
- **Database Action:** Order status updates to `Approved`. Logistics record transitions to `In Transit`.
- **Business Value:** Automated tracking keeps both FPO and SHG aligned on exact dispatch windows.

Step 4: Processing, Packing, & Traceability (SHG Cell)

- **Action:** Upon delivery, the SHG registers the raw inventory in their processing center and inputs packaging figures (e.g. crushing mustard seeds into branded bottles).
- **System Action:** Sourced materials decrement, and finished SHG consumer goods increment in the `shg\_inventories` collection.
- **Business Value:** Multiplies product margin, turning standard agricultural yield into higher-valued retail commodities.

Step 5: Buyer Connection & WhatsApp Fleet Notifications

- **Action:** The SHG uses the Proximity B2B Finder to match local demand. Once matched, they dispatch the order.
- **System Action:** System triggers an automated notification webhook, sending precise pickup locations, cost metrics, and cargo details directly to the transporter or buyer via WhatsApp.
- **Business Value:** Delivers instant operational feedback to local logistics networks without manual coordination.

Step 6: Compliance Audit & Submission Logger

- **Action:** The FPO clicks "Generate Audit Sheets" inside their Reports Center to download their digital compliance PDF.
- **Database Action:** The backend calculates totals dynamically and logs a unique PDF Hash Code (e.g., `FPC-3A8FD92E`) into MongoDB. The table dynamically prepends the new ledger row instantly.
- **Business Value:** Guaranteed audit trails and fast submission records for government agricultural departments.